

## NEW STUYAHOK, AK, USA

WMO: 702697

Lat: 59.450N

Lon: 157.333W

Elev: 302

StdP: 14.54

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00107

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-----|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |     |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |     |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p) |
| 12            | -15.3      | -11.2 | -22.6                         | 1.6 | -14.5 | -19.0 | 2.0 | -9.9 | 36.6                  | 33.6 | 33.9 | 35.2 | 11.1                  | 0    | N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.3                   | 73.9            | 59.3 | 69.8 | 57.5 | 66.1 | 55.8 | 60.9                | 71.0 | 58.9 | 66.9 | 57.2 | 63.8 | 8.5                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 57.2                            | 70.6 | 63.1 | 55.4 | 66.1 | 59.9 | 54.2 | 63.1 | 58.5 | 27.0          | 71.7 | 25.6 | 67.0 | 24.6 | 64.0 | 70.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 27.0          | 23.9 | 20.4 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |
| (5)               |      |      | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 34.2        | 17.0  | 17.8  | 20.4  | 31.1  | 43.7  | 51.3  | 55.3  | 54.9  | 47.5  | 35.5  | 19.4  | 15.7  |
|                                                                        | DBStd       | 18.21       | 14.73 | 14.00 | 12.21 | 8.70  | 5.94  | 5.35  | 5.17  | 3.99  | 5.79  | 8.80  | 12.10 | 14.80 |
|                                                                        | HDD50       | 6217        | 1024  | 901   | 917   | 566   | 213   | 39    | 9     | 6     | 110   | 450   | 919   | 1063  |
|                                                                        | HDD65       | 11239       | 1489  | 1321  | 1382  | 1016  | 662   | 413   | 305   | 313   | 526   | 915   | 1369  | 1528  |
|                                                                        | CDD50       | 460         | 0     | 0     | 0     | 0     | 16    | 79    | 173   | 158   | 34    | 0     | 0     | 0     |
|                                                                        | CDD65       | 8           | 0     | 0     | 0     | 0     | 0     | 3     | 5     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDH74       | 128         | 0     | 0     | 0     | 0     | 4     | 51    | 69    | 4     | 0     | 0     | 0     | 0     |
|                                                                        | CDH80       | 22          | 0     | 0     | 0     | 0     | 0     | 12    | 10    | 0     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 10.1        | 12.5  | 11.3  | 10.8  | 10.1  | 9.7   | 8.3   | 8.2   | 8.4   | 9.0   | 10.4  | 10.9  | 11.3  |
| (15) Precipitation                                                     | PrecAvg     | 32.5        | 2.4   | 2.1   | 1.9   | 1.7   | 1.9   | 2.2   | 3.0   | 3.3   | 4.3   | 3.5   | 3.1   | 3.1   |
|                                                                        | PrecMax     | 41.1        | 6.0   | 3.8   | 3.5   | 2.7   | 3.9   | 3.3   | 4.8   | 5.6   | 7.1   | 6.1   | 7.2   | 5.8   |
|                                                                        | PrecMin     | 24.6        | 0.7   | 0.5   | 0.4   | 0.2   | 0.7   | 0.7   | 1.5   | 1.1   | 1.8   | 0.5   | 0.4   | 1.4   |
|                                                                        | PrecStd     | 4.4         | 1.1   | 0.7   | 1.0   | 0.6   | 0.8   | 0.7   | 0.8   | 1.1   | 1.3   | 1.2   | 1.6   | 1.1   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 44.3  | 40.8  | 45.7  | 53.9  | 71.5  | 82.6  | 80.8  | 72.8  | 67.9  | 53.6  | 44.8  | 40.3  |
|                                                                        |             | MCWB        | 39.8  | 37.6  | 39.6  | 43.4  | 53.7  | 61.4  | 65.2  | 60.5  | 56.1  | 48.2  | 42.9  | 37.9  |
|                                                                        | 2%          | DB          | 39.8  | 38.3  | 40.9  | 48.6  | 64.4  | 73.3  | 74.9  | 69.5  | 61.5  | 50.9  | 41.3  | 38.2  |
|                                                                        |             | MCWB        | 38.0  | 35.7  | 37.2  | 40.4  | 50.8  | 57.2  | 60.9  | 58.9  | 53.1  | 46.6  | 40.3  | 36.0  |
|                                                                        | 5%          | DB          | 37.8  | 36.5  | 37.5  | 45.2  | 59.3  | 68.2  | 70.1  | 66.1  | 58.2  | 48.8  | 38.5  | 36.7  |
|                                                                        |             | MCWB        | 36.3  | 34.2  | 34.6  | 38.8  | 47.8  | 55.3  | 58.5  | 56.6  | 52.1  | 45.5  | 37.3  | 34.8  |
|                                                                        | 10%         | DB          | 36.2  | 34.5  | 35.6  | 41.9  | 54.5  | 63.3  | 66.1  | 63.4  | 55.4  | 46.9  | 35.8  | 35.2  |
|                                                                        |             | MCWB        | 34.8  | 32.7  | 33.1  | 37.0  | 45.4  | 52.5  | 56.8  | 55.4  | 51.1  | 44.5  | 34.9  | 34.0  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 40.3  | 38.0  | 40.1  | 44.0  | 56.7  | 62.1  | 66.6  | 62.8  | 57.3  | 49.9  | 43.6  | 38.6  |
|                                                                        |             | MCDB        | 43.5  | 39.8  | 44.0  | 51.9  | 69.4  | 80.1  | 80.0  | 68.9  | 62.9  | 51.3  | 45.0  | 39.6  |
|                                                                        | 2%          | WB          | 38.3  | 36.2  | 37.6  | 41.2  | 51.4  | 58.5  | 62.1  | 59.6  | 55.1  | 47.6  | 40.4  | 36.6  |
|                                                                        |             | MCDB        | 39.8  | 37.9  | 40.3  | 47.5  | 63.4  | 71.9  | 72.3  | 65.8  | 59.3  | 49.4  | 41.2  | 37.8  |
|                                                                        | 5%          | WB          | 36.6  | 34.4  | 35.4  | 39.3  | 48.3  | 55.7  | 59.6  | 58.2  | 53.5  | 46.3  | 37.4  | 35.1  |
|                                                                        |             | MCDB        | 37.7  | 36.0  | 37.5  | 44.1  | 57.0  | 66.7  | 68.4  | 63.9  | 56.7  | 48.1  | 38.2  | 36.3  |
|                                                                        | 10%         | WB          | 34.7  | 32.9  | 33.3  | 37.5  | 46.3  | 53.3  | 57.5  | 56.8  | 52.0  | 44.9  | 34.7  | 33.6  |
|                                                                        |             | MCDB        | 35.9  | 34.4  | 34.8  | 41.4  | 53.2  | 61.6  | 64.3  | 61.7  | 54.8  | 46.8  | 35.5  | 35.0  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 10.1  | 11.8  | 13.4  | 13.3  | 16.0  | 16.8  | 14.3  | 13.3  | 12.9  | 10.8  | 11.3  | 10.2  |
|                                                                        |             | MCDBR       | 8.6   | 9.4   | 12.1  | 16.1  | 25.0  | 26.2  | 23.6  | 19.2  | 16.5  | 10.7  | 11.5  | 10.3  |
|                                                                        | 5% WB       | MCWBR       | 7.6   | 8.0   | 9.0   | 10.1  | 13.5  | 12.5  | 11.3  | 9.9   | 9.7   | 7.8   | 10.7  | 9.3   |
|                                                                        |             | MCWBR       | 8.6   | 9.4   | 11.3  | 14.4  | 21.7  | 24.3  | 21.3  | 15.7  | 13.3  | 9.2   | 10.8  | 9.7   |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.252 | 0.297 | 0.307 | 0.342 | 0.377 | 0.362 | 0.362 | 0.356 | 0.329 | 0.302 | 0.306 | 0.247 |
|                                                                        |             | taud        | 2.118 | 2.168 | 2.196 | 2.238 | 2.283 | 2.401 | 2.439 | 2.454 | 2.500 | 2.424 | 2.018 | 1.975 |
|                                                                        | Ebn at Noon | Ebn at Noon | 171   | 222   | 261   | 272   | 269   | 276   | 273   | 266   | 256   | 226   | 133   | 116   |
|                                                                        |             | Edh at Noon | 14    | 24    | 31    | 36    | 37    | 34    | 32    | 29    | 24    | 19    | 16    | 11    |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 131   | 343   | 890   | 1351  | 1590  | 1692  | 1386  | 1141  | 805   | 418   | 173   | 77    |
|                                                                        | RadStd      | RadStd      | 18    | 47    | 120   | 140   | 176   | 168   | 152   | 135   | 107   | 44    | 27    | 9     |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (8 neighbors) | N/A   | +5.09   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page